

MRPL MANGALORE REFINERY

Plant Specification and Beginner-Friendly Factory Map

Purpose: Create a reliable plant specification for building an MRPL-informed DWSIM simulation and synthetic industrial dataset.

Research status: Public-source reconstruction. This report does not claim access to MRPL proprietary DCS / historian data, confidential P&IDs, proprietary kinetics, or exact control logic.

Prepared from public MRPL, Government of India and related engineering documents. Information current through publicly available FY 2025-26 results and April 2026 capacity data.

1. Executive summary

MRPL (Mangalore Refinery and Petrochemicals Limited) is a large coastal petroleum refinery at Kuthethoor/Katipalla, Mangaluru, Karnataka. The current PPAC table lists 15.0 million tonnes per year of installed refinery capacity as of 1 April 2026. MRPL reported 17.00 MMT of throughput (crude + other) for FY 2025-26, while FY 2024-25 gross crude throughput was reported at about 18.04 MMT. These numbers look inconsistent until you notice that MRPL has operated above the simple 15 MTPA nameplate basis and its environmental documents use operating-hour based capacity concepts. For a simulation, the safest choice is to use 15 MTPA as the nominal base and create higher-throughput scenarios separately.

The factory is not one machine. It is a network of process units, utilities and offsite systems. The main chain is: crude receipt -> crude/vacuum distillation -> heavy-oil conversion -> hydrotreating / upgrading -> product finishing -> blending / storage -> dispatch. Around that chain sit power, steam, hydrogen, water, cooling, flare, wastewater treatment, sulfur recovery and logistics.

For your AI project, the important point is that public sources are strong enough to define the **plant topology and unit roles**, but not strong enough to reproduce a proprietary high-frequency refinery historian. Therefore, DWSIM should generate the high-frequency synthetic process data, while public MRPL documents provide the real-world structure and constraints.

Item	Current / selected public fact
Company	Mangalore Refinery and Petrochemicals Limited (MRPL), subsidiary of ONGC
Location	Kuthethoor / Katipalla, Mangaluru, Dakshina Kannada, Karnataka
Current installed refinery capacity	15.0 MMTPA on PPAC table dated 1 Apr 2026 [1]
FY 2025-26 throughput	17.00 MMT throughput (crude + other) [2]
FY 2024-25 gross crude throughput	18.04 MMT, 120% capacity utilization stated by MRPL [3]
Site area shown in environmental compliance record	1,592 ha [public MRPL compliance record]
Main public expansion basis	2023 EC describes 16.6 -> 18.2 MMTPA by increasing operating hours from 8,000 to 8,760, without new units [4]
Refinery product family	LPG, naphtha, petrol/MS, HSD/diesel, kerosene, ATF, bitumen, sulfur, petcoke, etc. [6]
Polypropylene	440 KTA Novolen gas-phase plant in current MRPL brochure [6]

2. What is the whole MRPL factory?

Think of MRPL as a city whose roads are pipes. The process units are the factories inside the city, tanks are warehouses, utilities are the power and water departments, and the control system is the traffic-control center. A refinery works only when all of them cooperate.

The factory in one picture

1. CRUDE RECEIVING SPM / port -> crude storage	2. PRIMARY DISTILLATION CDU / VDU -> LPG, naphtha, kerosene, diesel, VGO, residue
3. CONVERSION HCU + DCU + PFCCU -> convert heavy material into more useful products	4. CLEAN-UP / UPGRADING HDT, GOHDS, DHDT, CCR, ISOM, Merox, SRU
5. PETROCHEMICALS PFCC propylene -> PPU polypropylene; aromatics -> para-xylene / benzene	6. OFFSITES + DISPATCH tankage, utilities, wastewater, jetties, pipelines, rail, marketing terminals

Beginner mental model: crude enters as a complicated mixture. The refinery first separates it, then converts the heavy parts, then removes impurities, then blends and stores finished products. Steam, hydrogen, cooling water, electricity, fuel gas, flare, wastewater treatment and safety systems keep that chain running.

Physical footprint around the main refinery

Public MRPL material shows several connected locations and assets. The main refinery is at Kuthethoor/Katipalla. Crude can come through MRPL's offshore Single Point Mooring (SPM) near Mangalore, and the company has associated pumping, pipeline, storage and jetty systems. MRPL also has a Sarapady pumping station that pumps raw water from the Nethravathi River to the refinery. Its desalination plant is at Tannirbhavi on the New Mangalore Port Authority area. The Aromatic Complex is in the Mangalore Special Economic Zone (MSEZ), Permude, and became part of MRPL through the OMPL amalgamation. These should be represented as connected facilities, not as if all equipment sits inside one fenced plot.

Facility / location	Simple job	Use in DWSIM project
Main refinery - Kuthethoor / Katipalla	Main crude processing and conversion	Core flowsheet
Offshore SPM	Receives large crude ships and transfers crude to shore system	Crude receiving boundary condition
Sarapady pumping station	Moves raw river water to refinery; no manufacturing there	Water utility source
Tannirbhavi desalination plant	Makes desalinated water for refinery needs	Water utility scenario
Aromatic Complex - MSEZ / Permude	Aromatics and associated products such as para-xylene and benzene	Separate downstream block; optional in v1
Marketing / dispatch infrastructure	Moves finished products by pipeline, jetty, rail and road	Offsite logistics / event layer

3. Main process units - beginner explanation

The table below uses the capacity figures in the January 2023 Environmental Clearance because it provides a single public unit-capacity table. The star (*) means the number is from that regulatory expansion basis, not necessarily a current measured operating rate. The current PPAC refinery capacity is still listed as 15.0 MMTPA [1].

Unit	What it does in simple language	Main feed	Public capacity* (MMTPA)
CDU/VDU (incl. NSU)	Separates crude into useful fractions. VDU further separates heavy residue under vacuum.	Crude	18.2*
VCU	Thermally cracks short residue to make lighter, lower-viscosity material.	Short residue	0.02*
BBU	Makes bitumen; the newer train was commissioned in Nov 2024.	Bitumen feed	0.19* total product basis
DCU	Converts heavy residue into lighter products and petroleum coke.	Short residue	3.48*
CHTU	Hydrotreats coker gas oil so it can be used as cleaner PFCC feed.	HCGO / VGO	1.01*
HCU	Uses hydrogen to crack heavy VGO into more valuable middle distillates.	VGO + H2	3.79*
PFCCU	Cracks heavy feed to make propylene, LPG, gasoline and light cycle oil.	VGO + UCO + treated HCGO	2.74*
PPU	Turns propylene into polypropylene pellets.	Propylene	0.51* / 440 KTA brochure basis
GOHDS + DHDT	Removes sulfur and other impurities from gas oils and diesel streams.	LGO/HGO/VGO or diesel streams	5.91*

3. Main process units - beginner explanation (continued)

Unit	What it does in simple language	Main feed	Public capacity* (MMTPA)
NHT / CCR	Treats naphtha and upgrades lower-octane naphtha into high-octane reformate.	Heavy naphtha	1.06*
RSU	Separates / handles reformate-related streams in the gasoline / aromatics chain.	Reformate-related streams	0.94*
Mixed Xylene	Handles aromatic xylene-rich material in the downstream chain.	Aromatic streams	0.56*
ISOM	Rearranges light naphtha molecules to improve octane.	Light naphtha	0.59*
HYD (DHDT)	Hydrotreating / hydrogen service associated with diesel treating.	Hydrocarbon stream + H ₂	0.12*
SRU	Recovers elemental sulfur from sulfur-bearing gas streams.	Sulfur-bearing gas	0.32*
KeroMerox	Treats kerosene-type stream to improve product quality.	Kerosene	1.10*
LPG Merox	Treats LPG to remove / convert sulfur compounds for product quality.	LPG	0.25*

4. How the main refinery process works

A. Crude receiving

Crude oil arrives by ship. The SPM allows larger crude carriers to unload offshore, and shore / pipeline / tank systems move the crude into storage. [7]

B. Crude distillation

CDU heats the crude and separates it into fractions mainly by boiling range. The lighter parts leave higher in the column; heavy material goes downward. [5]

C. Vacuum distillation

The heavy bottom from CDU is distilled under lower pressure. This lets MRPL recover valuable vacuum gas oils without requiring the same extreme temperatures that would cause more cracking. [5]

D. Conversion of heavy material

HCU uses hydrogen and catalysts to convert heavy VGO into more valuable middle distillates. DCU converts heavy residue into lighter products and petroleum coke. PFCC converts suitable heavy feeds into propylene, LPG, gasoline and light cycle oil. [5]

E. Hydrotreating / cleaning

GOHDS, DHDT and CHTU use hydrogen and catalysts to remove sulfur and other impurities and to prepare cleaner downstream feeds. [5]

F. Gasoline and naphtha upgrading

NHT/CCR and ISOM upgrade naphtha streams to improve octane and produce reformate / gasoline-range components. [5]

G. Product finishing

Merox units treat LPG / kerosene-type streams. Sulfur recovery converts sulfur-bearing gas into recoverable sulfur. Bitumen trains make paving-grade bitumen. [4],[5]

H. Petrochemicals

Propylene from PFCC can feed the polypropylene unit. MRPL also operates the Aromatic Complex for para-xylene and benzene. [6],[10]

5. Public product capacities used as the simulation envelope

The same 2023 environmental-clearance table lists the following refinery product pattern. These are best treated as **maximum / regulatory envelope values for the model**, not as a fixed product recipe. Actual yield changes with crude quality, unit operating mode, maintenance, market demand and optimization.

Product	Public capacity after the 2023 EC basis (MMTPA)
LPG	1.26*
Naphtha	1.771*
Motor Spirit / Petrol	1.96*
Kerosene	0.05*
ATF	2.04*
Diesel / HSD	6.47*
Fuel Oil	0.37*
Bitumen	0.19*
Sulfur	0.31*
Mixed Xylene	0.002*

Important additional product / petrochemical notes

MRPL's current brochure lists a broader product portfolio including LPG, polypropylene, naphtha, motor spirit, HSD, kerosene, ATF, mineral turpentine oil, reformat, xylol, bitumen, sulfur and petcoke. It also describes a 440 KTA Novolen gas-phase polypropylene facility. [6] The 2023 EC unit table shows 0.51 MMTPA for PPU, so your model should store both values with their source metadata rather than silently choosing one.

The Aromatic Complex was originally an OMPL project. Public MRPL material states the amalgamation of OMPL into MRPL became effective in May 2022, bringing the aromatic complex into the MRPL corporate structure. Older official material describes the complex as designed around roughly 914 KTPA para-xylene and 283 KTPA benzene. Use those as legacy design references, not as a claim about current live production.

6. Utilities - the systems that make the refinery possible

A refinery is not just reactors and columns. It consumes and produces large amounts of steam, electricity, hydrogen, fuel gas, cooling water and treated water. If you ignore utilities in DWSIM, your refinery will behave like a videogame that never runs out of electricity or water.

Utility / support system	What it does	What to simulate
Captive power + grid	Provides electricity to pumps, compressors, controls and other loads. MRPL has multiple captive power blocks and has been upgrading grid connectivity. [10]	Electrical load, availability, trips, import/export power
Steam system	Supplies process heating, stripping, turbines and other users.	Steam generation, headers, pressure levels, demand and imbalance
Hydrogen generation	Supplies hydrogen needed for hydroprocessing. CAG documented multiple HGUs historically. [5]	H2 production, purity, consumption, shortage events
Fuel gas / fired heaters	Provides heat to distillation and other process units.	Fuel rate, heater duty, efficiency, trip
Cooling water	Removes heat from condensers and coolers.	Supply temperature, return temperature, flow, exchanger duty
Raw / treated / desalinated water	Provides refinery water. Sarapady pumps Nethravathi river water; Tannirbhavi desalination reduces dependence on river water. [9],[10]	Water balance, source switching, supply interruption
Wastewater / ETP	Treats refinery wastewater before reuse / discharge.	Flow, contaminant load, treatment efficiency, recycle
Flare system	Safely burns excess / relieved hydrocarbon gases.	Flare load, relief event, gas composition
Sulfur recovery	Recovers sulfur from sour / acid gas streams.	Sour gas flow, sulfur production, recovery efficiency

Current water and energy facts worth preserving

MRPL expanded its Tannirbhavi desalination plant from 30 MLD to 40 MLD, with the project commissioned in February 2024. In FY 2024-25, MRPL reported consuming more than 5.6 million cubic meters of desalinated water and more than 5.2 million cubic meters of treated city sewage water, while recycling around 60% of internally generated effluent. [10]

MRPL also reports RLNG import capability of 0.8 MMSCMD, with natural gas used in the Hydrogen Generation Unit and as fuel for gas turbines. Its FY 2023-24 report records solar capacity of 6.06 MW at the refinery complex, 2.5 MW at the Aromatic Complex and 0.2 MW at the desalination unit. [10]

7. Crude in, product out: logistics around the factory

The plant is coastal. That is not decoration. Crude procurement and finished-product dispatch are part of the industrial system.

Asset	Beginner meaning	Model / data use
Single Point Mooring (SPM)	Floating / offshore crude receiving point for large ships. Public MRPL material places the facility about 17 km from the port area. [7]	Crude arrival events, ship size, unloading duration, transfer flow
Booster pumping / shore pipeline	Moves crude from marine transfer system toward refinery storage. [7]	Pipeline pressure, pump state, transfer rate
Crude / product tankage	Temporary storage between ships, process units and customers.	Tank level, inventory, grade, contamination, switching events
New Mangalore Port jetties	Used for smaller-vessel receipt / product dispatch and interconnection with refinery / aromatic lines. [5],[10]	Dispatch / receipt events, throughput
Pet coke rail loading	Rail system supports evacuation of petcoke. [10]	Daily / monthly dispatch events
Pipelines / marketing terminals	Move finished MS, HSD and ATF into the distribution network. [10]	Dispatch events and product balances

8. What happens during a normal day

- Crude is scheduled and received, then held in tanks.
- CDU/VDU runs continuously and sends several fractions to different downstream units.
- Hydroprocessing units consume hydrogen and remove impurities.
- Heavy residue and gas oils are converted in DCU / HCU / PFCC depending on the process plan.
- Gasoline and other products are upgraded and treated to meet specifications.
- Utilities keep steam, power, cooling, water and hydrogen available.
- Samples, online instruments and laboratory tests check product and process quality.
- Finished products move to tanks and then to ship, road, rail or pipeline dispatch.

9. Final DWSIM-ready MRPL specification

This is the part to use as the design contract for your simulation. It intentionally separates **what MRPL publicly documents** from **what DWSIM must assume**.

Specification item	Freeze this for v1
Facility identity	MRPL Mangalore refinery, with connected SPM / water / aromatic / dispatch context
Nominal refinery basis	15.0 MMTPA installed capacity from current PPAC table [1]
High-utilization scenario	16.6 MMTPA baseline and 18.2 MMTPA high-throughput case based on the 2023 EC operating-hours concept [4]
Core units	CDU/VDU, HCU, DCU, PFCCU, CHTU, NHT/CCR, GOHDS/DHDT, SRU, PPU
Secondary units	VBU, BBU, RSU, ISOM, KeroMerox, LPG Merox, HYD(DHDT), product blending
Utilities	Steam, power, hydrogen, fuel gas, cooling water, water, wastewater, flare
Logistics	SPM, crude pipeline / pumping, tanks, jetties, rail / road product movement
Aromatics	Separate connected downstream block; do not force a detailed aromatic flowsheet into v1 unless needed
Time-series sampling	10 seconds for critical dynamic assets; 1 minute for unit process data; 1 hour for plant KPIs
Primary AI asset	Recycle Gas Compressor / compressor train because MRPL publicly describes predictive-maintenance work in this area [3],[10]

Recommended first DWSIM flowsheet

Start with one coherent chain rather than the whole refinery:

Crude -> CDU/VDU -> VGO -> HCU -> UCO/HCU bottoms -> PFCC -> propylene -> PPU

At the same time: CDU/VDU heavy residue -> DCU -> coke + liquid products

And in parallel: naphtha -> NHT/CCR / ISOM -> gasoline-range streams

Then attach support systems: hydrogen + steam + cooling water + power + wastewater + flare.

Simulation scenarios to generate

- Normal operation at low / medium / high throughput
- Light, medium and heavy crude cases
- Sweet and sour crude cases
- Feed composition disturbance
- CDU furnace duty change
- Column pressure disturbance
- HCU hydrogen shortage
- Compressor efficiency degradation
- Compressor trip / restart
- Valve bias or stuck valve
- Heat-exchanger fouling
- Sensor drift, spike and dropout
- Utility power loss / load shedding
- Cooling-water temperature or flow disturbance
- Startup / shutdown / recycle mode

10. Dataset structure to generate from DWSIM

File / table	Contains	Example fields
facility.parquet	Plant identity and major sites	facility_id, location, site_type
units.parquet	Process units	unit_id, unit_type, capacity, feed, products
equipment.parquet	Major equipment	equipment_id, equipment_type, unit_id, duty
streams.parquet	Process streams	stream_id, source, destination, flow, T, P, composition
observations.parquet	High-frequency synthetic process data	timestamp, tag, value, unit, quality
events.parquet	Faults / alarms / operating events	event_id, start, end, asset, event_type, severity
features.parquet	Derived engineering features	pressure_ratio, efficiency, delta_T, rate_of_change
provenance.parquet	How each row was produced	run_id, scenario_id, DWSIM_version, seed, source_basis
graph_nodes.parquet	Knowledge graph nodes	node_id, type, name, source
graph_edges.parquet	Knowledge graph edges	subject, relation, object, evidence

Every synthetic observation should carry a source label

Use these labels so nobody mistakes simulation for proprietary MRPL history:

MRPL_PRIMARY_DOCUMENTED = directly stated in MRPL documents; GOVERNMENT_MRPL = government record about MRPL; INFERRED_STRUCTURE = engineering interpretation from multiple public sources; SYNTHETIC_DWSIM = generated by simulation; DERIVED_FEATURE = calculated from synthetic or public observations.

11. What we still do not know publicly

Unknown / proprietary area	Correct treatment
Actual DCS tag names and historian values	Do not invent. Generate synthetic tags.
Exact current PFD / P&ID; revision for every unit	Use public flow descriptions and mark topology as public-source reconstructed.
Exact catalyst kinetics / catalyst age	Use generic / literature-supported kinetics and mark assumptions.
Exact equipment datasheets / curves	Use public tender / generic engineering values only where needed.
Exact compressor vibration behavior	Use a degradation / fault model and label the output as a vibration proxy.
Exact product recipe at each moment	Let DWSIM determine outputs from scenario conditions; do not hard-code a daily recipe.
Exact current capacity interpretation above 15 MMTPA	Use 15 MMTPA as nominal current installed basis and 16.6 / 18.2 as separate high-utilization scenarios unless a newer regulatory capacity document is found.

12. Final interpretation for your project

You now have enough public information to define a **credible MRPL-informed refinery architecture**. You do not have enough information to claim an exact digital twin of MRPL. Those are different things.

For your research, that distinction is actually useful: the public documents become the evidence layer; DWSIM becomes the controlled simulator; your generated time series become the experimental dataset; and the AI system can be evaluated on anomaly detection, temporal reasoning, retrieval and predictive-maintenance tasks without pretending that proprietary refinery data were somehow leaked to you by the benevolent gods of GitHub.

The architecture already planned in your project is consistent with this model: separate semantic knowledge from numerical time series, preserve evidence / provenance, and build the retrieval baseline before fine-tuning. [project roadmap]

Appendix A - Sources used

These sources were selected because they directly support plant identity, process configuration, unit roles, utilities, logistics and recent operating status. Dates and values should be treated in the context of the source document year.

[1] PPAC - Installed Refining Capacity (current table dated 1 Apr 2026)

<https://ppac.gov.in/infrastructure/installed-refinery-capacity>

[2] MRPL - FY 2025-26 audited-results press release (24 Apr 2026)

<https://admin.mrpl.co.in/img/UploadedFiles/StatutoryDisclosure/Files/English/ca296ec1fda84dd2a34513895d4a3e50.pdf>

[3] MRPL - FY 2024-25 AGM / Chairman speech

<https://admin.mrpl.co.in/img/UploadedFiles/StatutoryDisclosure/Files/English/669836438160435fa0db346d0d86a4a3.pdf>

[4] MRPL - 2023 Environmental Clearance: expansion from 16.6 to 18.2 MMTPA

<https://mrpl.co.in/sites/default/files/10.%20EC%20for%20Cap%20Ex%2016.6%20to%2018.2%20MMTPA%20%289.1.2023%29.pdf>

[5] CAG Report No. 33 of 2017 - Planning, operation and support facilities of MRPL Phase III

<https://admin.mrpl.co.in/img/UploadedFiles/Report/Files/English/05918f4453044c72b84bbe958148e72d.pdf>

[6] MRPL - current refinery / polypropylene brochure

<https://mrpl.co.in/Content/Brochure.pdf>

[7] MRPL - 2024 Single Point Mooring Oil Spill Contingency Plan

<https://mrpl.co.in/sites/default/files/1.%20OSCP%20for%20SPM.pdf>

[8] MRPL - FY 2023-24 Environmental Statement for Aromatic Complex

<https://mrpl.co.in/sites/default/files/2.%20Env%20Statement%20ARC%20FY%202023-24.pdf>

[9] MRPL - FY 2023-24 Environmental Statement for Sarapady Pumping Station

<https://mrpl.co.in/sites/default/files/6.%20Env%20Statement%20Sarapady%20FY%202023-24.pdf>

[10] MRPL - FY 2023-24 Annual Report

<https://admin.mrpl.co.in/img/UploadedFiles/AnnualReport/Files/abbcad37c9544132a539b5cd027fc34f.pdf>

[11] DWSIM - official download

<https://dwsim.org/index.php/download/>

Appendix B - Terminology cheat sheet

Term	Full name	Meaning for a beginner
CDU	Crude Distillation Unit	First major separation step.
VDU	Vacuum Distillation Unit	Separates heavy material under vacuum.
HCU	Hydrocracker Unit	Uses hydrogen to convert heavy molecules into lighter products.
DCU	Delayed Coker Unit	Turns heavy residue into lighter products and petcoke.
PFCCU	Petrochemical Fluidized Catalytic Cracking Unit	Makes propylene, LPG, gasoline and other products from suitable heavy feeds.
CHTU	Coker Heavy Gas Oil Hydrotreating Unit	Cleans coker gas oil for downstream use.
DHDT	Diesel Hydro Desulphurisation Treating Unit	Removes sulfur and other impurities from diesel-related streams.
GOHDS	Gas Oil Hydro De-Sulphurisation	Hydrotreating of gas oils, including VGO-related streams.
CCR	Continuous Catalytic Reforming	Raises naphtha octane and makes reformat.
HGU	Hydrogen Generation Unit	Produces hydrogen for hydroprocessing.
SRU	Sulfur Recovery Unit	Recovers elemental sulfur.
PPU	Polypropylene Unit	Makes polypropylene from propylene.
SPM	Single Point Mooring	Offshore crude receiving system for large ships.
MMTPA	Million Metric Tonnes Per Annum	Capacity unit for large refinery throughput.
KTA / KTPA	Thousand Tonnes Per Annum	Capacity unit smaller than MMTPA.